Assignment 1 - Git & GitHub Fundamentals

Report

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Repository: github.com/Nour-2101/git-session1-NourElHouda

1. Repository Setup

I started with the website project and ran `git init -b main`. The first `git status` confirmed that the repository was on `main` with no commits yet. I added a `.gitignore` containing `local.env`, `*.log`, `!important.log`, and `Screenshots`. This ignores the local environment file, ordinary logs, and the screenshots folder while allowing `important.log` to be tracked. Before committing, I checked the staged filenames with `git diff --cached --name-only`.

Evidence: Figures 1 and 2.

2. README & Commit History

The README describes my small HTML, CSS, and JavaScript website, explains how to open `index.html`, and includes a GitHub profile link, a project image, and learning goals. I kept documentation and styling changes in separate commits, including `Initialize starter project with README and ignore rules` (`9b67fbb`), `Document Git learning goals` (`d9fcae5`), and `Add rounded` (`039fe44`). My history also contains `second commit` (`bb767e2`), which could have a more specific message.

Evidence: Figures 3, 4 and 5.

3. Selective Staging

I changed `README.md` and `style.css`, but staged only the README first with `git add README.md`. `git status --short` then showed the README staged and `style.css` still modified but unstaged. I committed the README as `Document Git learning goals`; afterward, `style.css` was still listed as modified. I then staged and committed the CSS change separately as `Add rounded`.

Evidence: Figures 5, 6 and 7.

4. Branch & Merge

I created the `feature/about` branch with `git switch -c feature/about` and verified the active branch using `git branch --show-current`. I later switched back to `main`, but the screenshots do not show a completed merge. My `git merge --no-ff` attempt did not specify a branch and returned an error; `index.html` still had uncommitted changes. The displayed history shows `main` and `feature/about` pointing to the same commit. I still need to commit the feature work and merge `feature/about` into `main` before reporting this step as finished.

Evidence: Figures 8 and 9.

5. Remote Changes

A `commit` saves a version of my changes in the local repository; a `push` sends commits to GitHub. I added the `origin` remote and ran `git push -u origin main`, which pushed the branch and set it to track `origin/main`.

Later, a GitHub commit called `Add progress section to README` was available remotely. `git fetch origin` updated the remote-tracking branch without changing my local working files, and `git status` showed that `main` was one commit behind. Then `git pull --ff-only origin main` fast-forwarded my local branch. After the pull, `main` was up to date with `origin/main`, although `index.html` still had a local unstaged modification.

Evidence: Figures 10, 11 and 12.

6. Fair Use of AI

I used AI assistance to organize and write this report from my screenshots and repository history. The screenshots are the evidence of the Git commands I ran. Where the available output does not confirm a successful step, such as the feature merge, I have described it as incomplete instead of claiming it worked.

7. Sources & Reflection

Sources: my Git Bash screenshots, the GitHub project repository linked above, and the Git documentation (`https://git-scm.com/docs`).

The hardest part was the branch and merge workflow because creating a branch is not the same as merging its changes. If Git behaves unexpectedly, I would first run `git status` to check my working files, then `git branch -vv` and `git log --oneline --graph --decorate --all` to see where the branches point.

Appendix: selected screenshots | Repository setup

```
DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git init -b main
Initialized empty Git repository in C:/Users/DELL-PC/Downloads/git-session1-Nour-2101/.git/

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app.js
        app.log
        assets/
        debug.log
        important.log
        index.html
        local.env
        notes.txt
        style.css

nothing added to commit but untracked files present (use "git add" to track)

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
```

Figure 1. Initialize the repository on main and inspect untracked files.

```
DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git add .gitignore README.md index.html style.css app.js notes.txt important.log assets/project-banner.svg
git diff --cached --name-only
warning: in the working copy of 'app.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'assets/project-banner.svg', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'important.log', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'style.css', LF will be replaced by CRLF the next time Git touches it
.gitignore
README.md
app.js
assets/project-banner.svg
important.log
index.html
notes.txt
style.css

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$
```

Figure 2. Review the staged files before the initial commit.

Appendix | README and commit history

Git & GitHub — Session 1 Assignment

Name: Nour El Houda Garaoui *GitHub username:* Nour-2101

Description

A small website used to practice Git repositories, staging, commits, branches, merging, and synchronization with GitHub.

Technologies

- HTML
- CSS
- JavaScript
- Git
- GitHub

Run the project

Open index.html in your browser.

Example command

Check the repository status:

bash git status

GitHub Profile

Visit my GitHub profile

Figure 3. README content: description, technologies and run instructions.

```
DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git commit -m "Initialize starter project with README and ignore rules"
[main (root-commit) 9b67fbb] Initialize starter project with README and ignore rules
 8 files changed, 61 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 app.js
 create mode 100644 assets/project-banner.svg
 create mode 100644 important.log
 create mode 100644 index.html
 create mode 100644 notes.txt
```

Figure 4. First commit records the project files and .gitignore.

```
On branch main
nothing to commit, working tree clean

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git log --oneline
039fe44 (HEAD -> main) Add rounded
d9fcae5 Document Git learning goals
bb767e2 second commit
9b67fbb Initialize starter project with README and ignore rules

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
```

Figure 5. Commit history includes separate README and CSS changes.

Appendix | Selective staging

```
DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git diff --cached
diff --git a/README.md b/README.md
index 4f9e640..d7d76f5 100644
--- a/README.md
+++ b/README.md
@@ -35,4 +35,11 @@ git status

 ## Project Image

-![Project banner](assets/project-banner.svg)
\ No newline at end of file
+![Project banner](assets/project-banner.svg)
+
+## Learning Goals
+
+- Record changes with meaningful commits.
+- Stage selected files independently.
+- Create and merge feature branches.
+- Synchronize a local repository with GitHub.
\ No newline at end of file

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git status --short
M  README.md
 M style.css

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$
```

Figure 6. README staged while style.css remains unstaged.

```
$ git commit -m "Document Git
> learning goals"
[main d9fcae5] Document Git learning goals
 1 file changed, 8 insertions(+), 1 deletion(-)

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-
2101 (main)
$ git status --short
 M style.css

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-
2101 (main)
$
```

Figure 7. README committed; the CSS change remains uncommitted.

Appendix | Feature branch and merge attempt

```
$ git switch -c feature/about

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-
2101 (feature/about)
$ git branch --show-current
feature/about

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-
2101 (feature/about)
$
```

Figure 8. Create and confirm the feature/about branch.

```
DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (feature/about)
$ git switch main
M       index.html
Switched to branch 'main'

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git merge --no-ff
fatal: No remote for the current branch.

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ feature/about -m "Merge
> About feature into main"
bash: feature/about: No such file or directory

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git log --oneline --graph --
* 039fe44 (HEAD -> main, feature/about) Add rounded
* d9fcae5 Document Git learning goals
* bb767e2 second commit
* 9b67fbb Initialize starter project with README and ignore rules

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
```

Figure 9. Merge attempt fails; index.html remains uncommitted.

Appendix | Push and fetch

```
DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git remote add origin https://github.com/Nour-2101/git-session1-NourElHouda.git

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git branch -M main

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git push -u origin main
Enumerating objects: 20, done.
Counting objects: 100% (20/20), done.
Delta compression using up to 6 threads
Compressing objects: 100% (16/16), done.
Writing objects: 100% (20/20), 2.72 KiB | 69.00 KiB/s, done.
Total 20 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
To https://github.com/Nour-2101/git-session1-NourElHouda.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$
```

Figure 10. Add origin and push main to GitHub.

```
DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git fetch origin
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1.01 KiB | 23.00 KiB/s, done.
From https://github.com/Nour-2101/git-session1-NourElHouda
   039fe44..d0947fc  main       -> origin/main

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
```

Figure 11. Fetch the remote change; local main is one commit behind.

Appendix | Pull and final status

```
DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git pull --ff-only origin main
From https://github.com/Nour-2101/git-session1-NourElHouda
 * branch            main       -> FETCH_HEAD
Updating 039fe44..d0947fc
Fast-forward
 README.md | 5 ++++-
 1 file changed, 4 insertions(+), 1 deletion(-)

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")

DELL-PC@DESKTOP-CRVBNCQ UCRT64 ~/Downloads/git-session1-Nour-2101 (main)
$ git log --oneline -3
d0947fc (HEAD -> main, origin/main, origin/HEAD) Add progress section to README
039fe44 (feature/about) Add rounded
d9fcae5 Document Git learning goals
```

Figure 12. Fast-forward main after fetching; index.html is still modified locally.